

CSCI-739 Quantum Machine Learning

Homework 1 — Expert-Assessment Version

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Basis convention: written kets use $|q_0q_1\cdots\rangle$; Qiskit count strings are mapped explicitly below. The comprehensive reference gives the extended derivations and verification notebook contains runnable circuits.

Problem 1. Qubit Basics

(10 pts)

Given: a general single-qubit state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, with $\alpha, \beta \in \mathbb{C}$ and $|\alpha|^2 + |\beta|^2 = 1$.

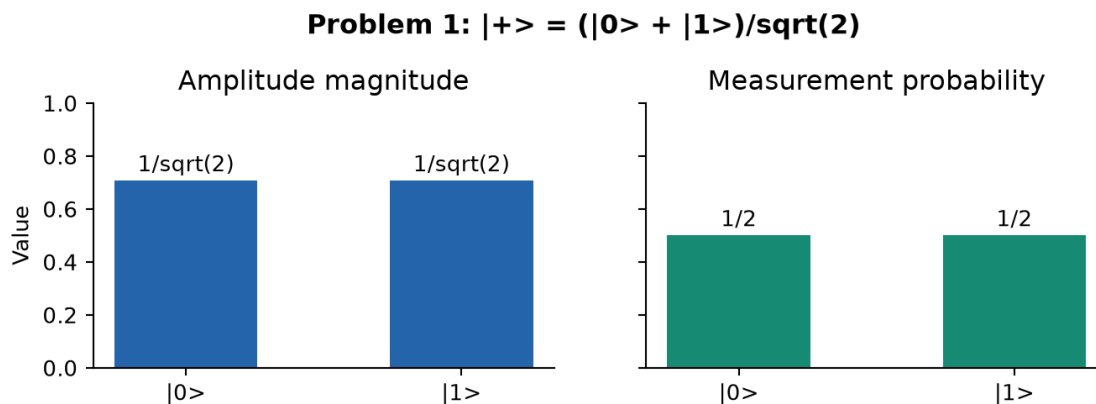


Figure 1: Equal-superposition amplitude magnitudes and measurement probabilities; the magnitude is squared to obtain probability.

- (a) **Question.** Write the expression for the probability of measuring the qubit in the state $|0\rangle$ and in the state $|1\rangle$, using inner products (bra-ket notation).
Answer. For $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, $\langle 0|\psi\rangle = \alpha$ and $\langle 1|\psi\rangle = \beta$; hence $P(0) = |\alpha|^2$ and $P(1) = |\beta|^2$.
- (b) **Question.** Verify that the sum of the two probabilities is always equal to 1.
Answer. The normalization condition gives $P(0) + P(1) = |\alpha|^2 + |\beta|^2 = 1$, equivalently $\langle\psi|\psi\rangle = 1$.
- (c) **Question.** For the specific state $|\psi\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$, calculate explicitly the probability of obtaining $|0\rangle$ and of obtaining $|1\rangle$.
Answer. Each amplitude has magnitude $1/\sqrt{2}$, so $P(0) = P(1) = 1/2$. One measurement returns one outcome; these probabilities describe repeated measurements.

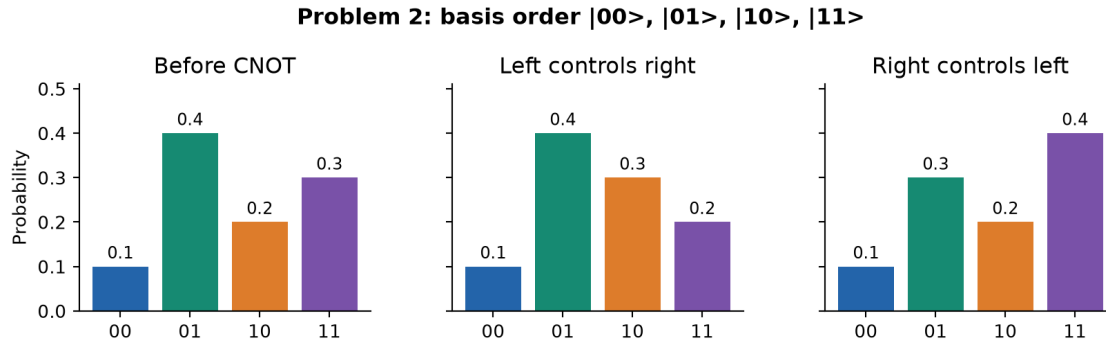


Figure 2: Exact computational-basis probabilities before CNOT and after each control direction.

- (a) **Question.** Define each of the four basis states $|00\rangle, |01\rangle, |10\rangle, |11\rangle$ explicitly using tensor product notation between single-qubit states, and then write the vector representation of each of the four basis states.

Answer. In order $(00, 01, 10, 11)$, the states are $|0\rangle \otimes |0\rangle, |0\rangle \otimes |1\rangle, |1\rangle \otimes |0\rangle, |1\rangle \otimes |1\rangle$; their vectors are $(1, 0, 0, 0)^T, (0, 1, 0, 0)^T, (0, 0, 1, 0)^T, (0, 0, 0, 1)^T$.

- (b) **Question.** Consider the two-qubit state $|\psi\rangle = \sqrt{\frac{1}{10}}|00\rangle + \sqrt{\frac{4}{10}}|01\rangle + \sqrt{\frac{2}{10}}|10\rangle + \sqrt{\frac{3}{10}}|11\rangle$. Verify normalization and state the probability of each basis outcome. *Note: In this example the amplitudes are real, but in general they are complex, and the probability is the modulus squared of the amplitude.*

Answer. The squared norm is $(1 + 4 + 2 + 3)/10 = 1$; probabilities in the stated order are $(1, 4, 2, 3)/10$.

- (c) **Question.** Compute the probability of measuring each basis state after applying a CNOT gate with the left qubit as control and the right as target.

Answer. The map is $00 \rightarrow 00, 01 \rightarrow 01, 10 \rightarrow 11, 11 \rightarrow 10$. It exchanges the last two amplitudes, so the probabilities become $(1, 4, 3, 2)/10$.

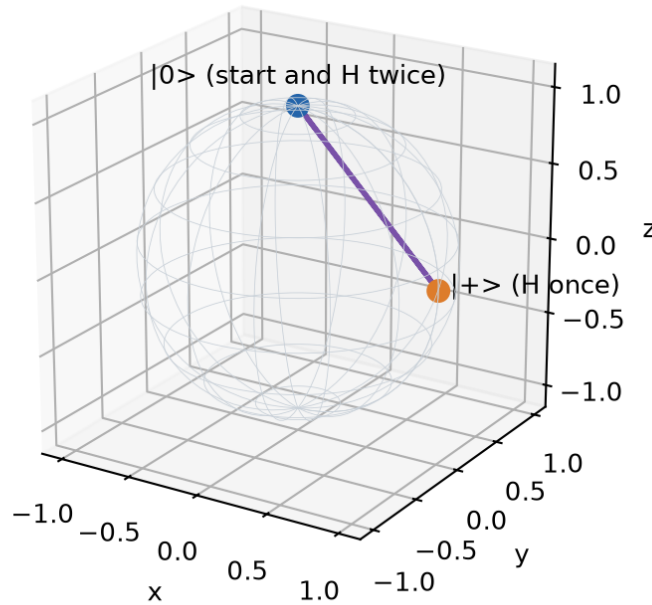
- (d) **Question.** Derive the 4×4 unitary matrix of the flipped-control CNOT (right controls, left is target). *Hint: track where each basis state maps under this assumption, and create a unitary matrix that would map each basis state to the expected operation.*

Answer. Since $|ab\rangle \mapsto |a \oplus b, b\rangle$, the map is $00 \rightarrow 00, 01 \rightarrow 11, 10 \rightarrow 10, 11 \rightarrow 01$, giving

$$U_{R \rightarrow L} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}.$$

- (e) **Question.** Prove that the matrix in (d) is unitary.

Answer. This real symmetric matrix swaps $01 \leftrightarrow 11$ and fixes the other basis states. Thus $U^\dagger = U, U^2 = I_4$, and $U^\dagger U = I_4$.

Problem 3: H moves $|0\rangle$ to $|+\rangle$ and back

 Figure 3: Bloch-sphere positions in the sequence $|0\rangle, H|0\rangle=|+\rangle, H^2|0\rangle=|0\rangle$.

- (a) **Question.** Any single-qubit state can be written as $|\psi\rangle = \cos(\theta/2)|0\rangle + e^{i\phi}\sin(\theta/2)|1\rangle$, where $\theta \in [0, \pi]$, $\phi \in [0, 2\pi)$. Derive θ and ϕ in terms of amplitudes α, β for $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$. State (θ, ϕ) for $|0\rangle$ and $|1\rangle$.

Answer. Remove the global phase of nonzero α ; then $\theta = 2\arctan2(|\beta|, |\alpha|)$ and $\phi = (\arg \beta - \arg \alpha) \bmod 2\pi$ when both amplitudes are nonzero. The poles are $|0\rangle : (0, \text{arbitrary})$ and $|1\rangle : (\pi, \text{arbitrary})$.

- (b) **Question.** For each quantum gate $X, Y, Z, H, P(\hat{\theta})$ ($\hat{\theta} \in [0, 2\pi)$ – this is sometimes referred to as the phase gate), describe qualitatively how each transforms (α, β) , (θ, ϕ) , and their impact on measurement of the basis states after their application relative to measurements of the original quantum state.

Answer. Writing $\mathbf{r} = (\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta)$, the transformations are:

Gate	Amplitudes	Bloch action	Z probabilities
X	(β, α)	$(r_x, -r_y, -r_z); (\pi - \theta, -\phi)$	$(\beta ^2, \alpha ^2)$
Y	$(-i\beta, i\alpha)$	$(-r_x, r_y, -r_z); (\pi - \theta, \pi - \phi)$	$(\beta ^2, \alpha ^2)$
Z	$(\alpha, -\beta)$	$(-r_x, -r_y, r_z); (\theta, \phi + \pi)$	unchanged
$P(\hat{\theta})$	$(\alpha, e^{i\hat{\theta}}\beta)$	z rotation; $(\theta, \phi + \hat{\theta})$	unchanged
H	$((\alpha + \beta)/\sqrt{2}, (\alpha - \beta)/\sqrt{2})$	$(r_z, -r_y, r_x); \theta' = \arccos r_x, \phi' = \arctan2(-r_y, r_z)$	$(\frac{1}{2} + \operatorname{Re} \alpha^* \beta, \frac{1}{2} - \operatorname{Re} \alpha^* \beta)$

Angles are modulo 2π ; azimuth is arbitrary at the poles. Z and P can change relative phase without changing immediate Z probabilities; H can make that phase observable. A global phase changes no measurement.

- (c) **Question.** Apply H to $|0\rangle$, write the resulting state.

Answer. $H|0\rangle = (|0\rangle + |1\rangle)/\sqrt{2} = |+\rangle$.

- (d) **Question.** Apply H again, describe which basis state undergoes constructive/destructive interference and explain the outcome.

Answer. $H|+\rangle = \frac{1}{2}[(|0\rangle + |1\rangle) + (|0\rangle - |1\rangle)] = |0\rangle$. The $|0\rangle$ amplitudes reinforce and the $|1\rangle$ amplitudes cancel.

- (e) **Question.** Is the final state a superposition or a determined basis state? Justify.

Answer. It is the determined basis state $|0\rangle$, since $H^2 = I$.



Figure 4: Exact outcome distributions: part (d) applies H to q0 only; part (e) applies H to all three qubits.

- (a) **Question.** Prove that $(H \otimes H)|00\rangle = (H|0\rangle) \otimes (H|0\rangle)$. Explain why this generalizes to n qubits.

Answer. Since $H|0\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$, both sides equal $\frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$. Repeating the componentwise action proves $(\bigotimes_j U_j)(\bigotimes_j |\psi_j\rangle) = \bigotimes_j U_j |\psi_j\rangle$.

- (b) **Question.** Explain how tensoring unitaries allows us to define operations on multi-qubit systems.

Answer. Tensor products act on the joint Hilbert space, and $(U \otimes V)^\dagger (U \otimes V) = (U^\dagger U) \otimes (V^\dagger V) = I \otimes I$. Use identity operators on wires untouched at a circuit step; a joint CNOT acts branchwise.

- (c) **Question.** Construct a Qiskit circuit on two qubits applying H to both, and verify it matches your result in (a).

Answer. The notebook applies H to q_0, q_1 and checks the statevector $|++\rangle$. Each computational-basis result has probability $1/4$; the seeded 1024-shot counts were 00 : 257, 01 : 255, 10 : 266, 11 : 246.

- (d) **Question.** On three qubits: Case 1—apply H to q_0 only, then CNOT(0→1) (with convention being control qubit number → target qubit number, starting qubit indexing at 0 because we count our qubits like real computer scientists). Case 2—apply H to q_0 only, then CNOT(1→0). In both cases, show mathematically (tensor product or simulation) how you can do an analogous computation to what was done in (a) for more general multi-qubit systems. Hint: if nothing is done to a qubit at a step of the quantum circuit, use the identity matrix as the quantum operation on it (the do-nothing gate). See the circuits below to get an idea of each case, following Case 1 and then Case 2; the figures show H on every qubit, but you should apply H only to q_0 .

Answer. Before CNOT, $(H \otimes I \otimes I)|000\rangle = (|000\rangle + |100\rangle)/\sqrt{2}$. Case 1 gives $(|000\rangle + |110\rangle)/\sqrt{2}$; Case 2 leaves $(|000\rangle + |100\rangle)/\sqrt{2}$ unchanged because the control $q_1 = 0$. Seeded counts in Qiskit order $q_2 q_1 q_0$ were Case 1: 000 : 521, 011 : 503; Case 2: 000 : 521, 001 : 503.

- (e) **Question.** *Food for thought:* if you instead apply H to all three qubits before the CNOT (as in the figures), show that Case 1 and Case 2 produce exactly the same state. Why does the CNOT direction not matter here? (Hint: what is CNOT $|++\rangle$? Is $|++\rangle$ entangled?)

Answer. The state is the product $|++\rangle$. Since $X|+\rangle = |+\rangle$, either CNOT leaves it unchanged; the two gates remain different on other inputs. The statevectors were equivalent, and both circuits produced the same seeded counts: 000 : 139, 001 : 124, 010 : 127, 011 : 134, 100 : 131, 101 : 127, 110 : 124, 111 : 118, versus exact probability $1/8$ each.

Problem 5: full-adder truth table

000	0	0
001	1	0
010	1	0
011	0	1
100	1	0
101	0	1
110	0	1
111	1	1

Sum S Carry Cout

Figure 5: Full-adder outputs for every A, B, Cin input; the Cin=0 rows also check the half-adder outputs.

- (a) **Question.** Work out by hand what happens to $|110\rangle$ as input to the half quantum adder and $|1100\rangle$ for the full adder. What is the final expected quantum state before measurement?

Answer. The half-adder gates $CCX(0,1,2)$, $CX(0,1)$ give $|110\rangle \rightarrow |111\rangle \rightarrow \boxed{|101\rangle}$, so $S = 0, C = 1$. The full-adder gates are $CCX(0,1,3)$, $CX(0,1)$, $CCX(1,2,3)$, $CX(1,2)$, $CX(0,1)$, giving

$$|1100\rangle \rightarrow |1101\rangle \rightarrow |1001\rangle \rightarrow |1001\rangle \rightarrow |1001\rangle \rightarrow \boxed{|1101\rangle},$$

so $S = 0, C_{out} = 1$. These are full register states before measurement.

- (b) **Question.** Implement both circuits in Qiskit, initializing carry to $|0\rangle$ and setting inputs with X gates as needed.
Answer. The notebook follows the pinned gate order, initializes carry targets at zero, and prepares each input from $|0 \dots 0\rangle$ with conditional X gates.
- (c) **Question.** Run for inputs $(q_0, q_1) \in \{00, 01, 10, 11\}$ for the half adder. Do the same with the full adder, with $C_{in} = q_2 = 0$. Use conditional X gates to prepare inputs that start in the ground state on the appropriate qubit. Record outputs (q_1, q_2) which represent the value of the addition and if carry is present, respectively for the half adder, and measure (q_2, q_3) for the full adder, which hold S and C_{out} respectively.

Answer. Half outputs are $(S, C) = (q_1, q_2)$; full outputs are $(S, C_{out}) = (q_2, q_3)$. Since $S \rightarrow c_0$ and carry $\rightarrow c_1$, Qiskit prints CS :

Input AB	Expected (S,C)	Half counts	Full counts
00	(0,0)	00:1024	00:1024
01	(1,0)	01:1024	01:1024
10	(1,0)	01:1024	01:1024
11	(0,1)	10:1024	10:1024

- (d) **Question.** Compare the quantum adder results using a simulator did you find what you expected? You are in no way required to, but feel free to run this same circuit on Qiskit's real quantum computers instead of simulator - did your results change?

Answer. Ideal Aer (1024 shots, seed 739) matched all predicted outputs. With $C_{in} = 1$, inputs 001, 011, 101, 111 printed 01, 10, 10, 11, each 1024 times. No hardware run was performed; hardware is optional and may produce noisy distributions.

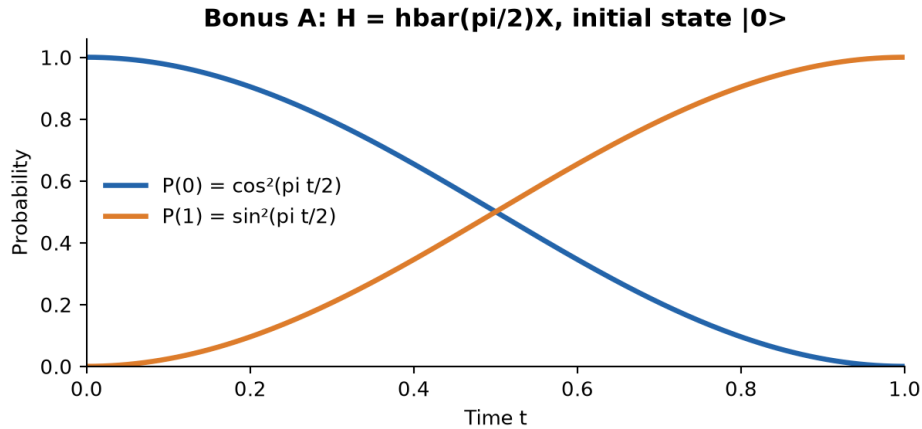


Figure 6: Example evolution from $|0\rangle$ under $H = \hbar(\pi/2)X$; the general amplitude-swap result is derived below.

Given. The time-dependent Schrödinger equation is $i\hbar d|\psi(t)\rangle/dt = H(t)|\psi(t)\rangle$, with $|\psi(0)\rangle = \alpha|0\rangle + \beta|1\rangle$, $\alpha, \beta \in \mathbb{C}$, and $|\alpha|^2 + |\beta|^2 = 1$. The Hamiltonian is Hermitian.

- (a) **Question.** Assume the Hamiltonian commutes with itself at different times, $[H(t), H(t')] = 0$ for all t, t' (this is automatic if H is time-independent). Consider the general solution to this equation under that assumption: $|\psi(t)\rangle = U(t)|\psi(0)\rangle$, $U(t) = e^{-\frac{i}{\hbar} \int_0^t H(\tau) d\tau}$, and prove it solves the Schrödinger equation. (Without this assumption the solution is a time-ordered exponential, which is beyond the scope of this problem.)

Answer. Let $A(t) = -(i/\hbar) \int_0^t H(\tau) d\tau$. Commutation gives $[A, A'] = 0$, so $U' = A'U = -(i/\hbar)HU$. Thus $i\hbar d(U|\psi(0)\rangle)/dt = HU|\psi(0)\rangle$, and $U(0) = I$ supplies the initial condition.

- (b) **Question.** Prove $U(t)$ is unitary if $H(t)$ is Hermitian. A matrix $U(t)$ is said to be unitary if $U^\dagger(t)U(t) = U(t)U^\dagger(t) = I$, where $U^\dagger(t)$ denotes the conjugate transpose (Hermitian adjoint) of $U(t)$. Unitarity guarantees that inner products and probabilities are preserved under time evolution. A matrix $H(t)$ is said to be Hermitian if $H^\dagger(t) = H(t)$. Hermiticity ensures that the eigenvalues of the Hamiltonian are real, so that measurable energies are real numbers.

Answer. Hermitian H makes $A^\dagger = -A$. Therefore $U^\dagger = e^{-A}$ and $U^\dagger U = U U^\dagger = I$.

- (c) **Question.** For $H(t) = \hbar\Omega \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ on $[0, 1]$, compute $U(t)$, and choose a real number Ω so that $U(1)$ swaps α, β (up to a global phase). Note Ω must be real so that $H(t)$ stays Hermitian.

Answer. Since $X^2 = I$,

$$U(t) = e^{-i\Omega t X} = \cos(\Omega t)I - i\sin(\Omega t)X = \begin{pmatrix} \cos \Omega t & -i\sin \Omega t \\ -i\sin \Omega t & \cos \Omega t \end{pmatrix}.$$

Choose $\Omega = \pi/2$, giving $U(1) = -iX$ and $|\psi(1)\rangle = -i(\beta|0\rangle + \alpha|1\rangle)$.

- (d) **Question.** Show $|\psi(1)\rangle$ is normalized, e.g. the sum of the probabilities equals 1.

Answer. The probabilities are $(|\beta|^2, |\alpha|^2)$, whose sum is $|\alpha|^2 + |\beta|^2 = 1$.

- (e) **Question.** Identify the effective one-qubit gate implemented from the time evolution of the quantum system over $t \in [0, 1]$.

Answer. It is Pauli X , up to the global phase $-i$.

Bonus B: Cout S1 S0 for all 2-bit pairs

a=0	0 000	1 001	2 010	3 011
a=1	1 001	2 010	3 011	4 100
a=2	2 010	3 011	4 100	5 101
a=3	3 011	4 100	5 101	6 110
	b=0	b=1	b=2	b=3

Figure 7: All 16 two-bit input pairs and their three-bit outputs Cout S1 S0; the orange box marks $2+2=4$.

- (a) **Question.** Implement a 2-bit ripple-carry adder with two 2-qubit registers (a_1a_0, b_1b_0) and carry c .
Answer. Map $(q_0, q_1, q_2, q_3, q_4, q_5) = (a_0, a_1, b_0, b_1, c_1, c_{out})$. The low half-adder uses CCX(0,2,4), CX(0,2); the high full adder uses CCX(1,3,5), CX(1,3), CCX(3,4,5), CX(3,4), CX(1,3). The low carry feeds the high stage.
- (b) **Question.** Prepare $a = 2$ and $b = 2$ in binary (10_2 each), with $c = 0$.
Answer. Starting from zero, apply X to q_1, q_3 . In written $q_0 \dots q_5$ order, the input is $|010100\rangle$; both carry/work qubits start at zero.
- (c) **Question.** For the correct binary inputs, work through the quantum circuit (either by hand or with computer) to work out the expected result $2 + 2 = 4 = 100_2$. Show how this appears as (s_0, s_1, c_{out}) on your qubits and as a bitstring in the Qiskit register.
Answer. The low stage gives $(s_0, c_1) = (0, 0)$; the high stage gives $(s_1, c_{out}) = (0, 1)$. The full register trace is $|010100\rangle \rightarrow |010101\rangle \rightarrow |010001\rangle \rightarrow |010001\rangle \rightarrow |010001\rangle \rightarrow |010101\rangle$. Thus $(s_0, s_1, c_{out}) = (0, 0, 1)$.
- (d) **Question.** Simulate and record the dominant measurement outcome.
Answer. Statevector checks match ordinary addition for all 16 two-bit input pairs. For this circuit the ideal 1024-shot Aer result (seed 739) is 100:1024, i.e. $100_2 = 4$.
- (e) **Question.** (Optional) Run on hardware (~ 100 shots). Compare simulator vs. noisy hardware outcomes.
Answer. No hardware run was performed; the assignment marks it optional.
- (f) **Question.** State clearly your measurement mapping for (s_0, s_1, c_{out}) . Hint: follow the important point I mentioned, so you really internalize the ordering of bits in a qubit versus binary representation/register ☺.
Answer. Measure $q_2 \rightarrow c_0 = s_0$, $q_4 \rightarrow c_1 = s_1$, and $q_5 \rightarrow c_2 = c_{out}$. Qiskit prints $c_2c_1c_0 = 100$, the reverse-index register string for these mapped output bits.

References by part: (a)–(f): QML0 / Lecture 1, slides 57–61 (circuit reading and qubit-order conventions).

AI-use statement: AI assistance was used to review my existing written and coded work, study the course lectures and notes, map and reference relevant material for each problem and bonus, and automate preparation of this document.